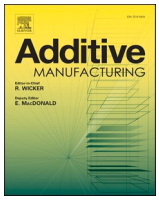
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Printability criterion and filler characteristics model for material extrusion additive manufacturing

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ABSTRACT

Material extrusion is an additive manufacturing technique that enables the creation of reproducible and complex hardware by depositing a viscous, shear-thinning ink onto a substrate in a custom-pattern via extrusion through a syringe. The ability of an ink to be extruded onto a substrate in many layers and maintain the desired shape is what defines printability. Printability has historically been investigated in an iterative manner by formulating and printing inks and then performing postmortem analysis of final parts. Highly concentrated pastes continue to pose issues for practitioners as the effect of filler morphology and size dispersity on the ink rheology and corresponding printability is not well understood. A printability criterion based on the particle filler's maximum packing fraction was recently proposed to provide a general framework to understand printability of particle-filled inks. Inks were found to be printable if the particle loading was within 90–94 % of the maximum packing fraction of the particle. Here we expand on that work to validate the generality of the maximum packing fraction criterion by testing with 10 new single and multimodal particle fillers. The maximum packing fraction calculated from small amplitude oscillatory shear experiments and is found to correctly predict the printability range for all inks. We then utilize statistical methods to develop a filler characteristics model to predict the maximum packing fraction from particle analysis alone. These two methods paired together can significantly speed up development of new inks, increase the performance of material extrusion printing, and improve the stability of printed parts, with less wasted time and materials.

1. Introduction

Additive Manufacturing (AM) provides the ability to produce complex parts that are impossible to obtain via traditional manufacturing techniques. Material extrusion through Direct Ink Write (DIW) is an AM technique which often involves extruding a viscous ink through a nozzle where material is deposited onto a substrate to create parts in a layer-by-layer fashion. Common DIW inks are suspensions of particle fillers, such as ceramics [1,2], metals [3,4], or organics [5,6], in a base, such as silicones [7], epoxies [8], or urethanes [9]. The applications of DIW range from building materials [10–13], energy storage [14,15], medical devices [16–18], geology [19–21], and biology [5,6]. In this work we focus on a subset of DIW AM inks, dense pastes with hard particle fillers. Specifically, the inks discussed here are those which interaction through frictional contacts rather than some surface charge or chemistry. One of the major challenges in this space of DIW AM is the development of characterization techniques to determine if new ink formulations will be printable, or more clearly, have the required characteristics to be extruded smoothly and hold the printed shape.

Current screening methods for DIW inks require new formulations to be tested on-line to determine if an ink will be printable. This often requires large quantities of ink, multiple lengthy printing processes, *in-situ* monitoring, or post-mortem analysis [22–24]. This has led many researchers to investigate benchtop characterization methods for screening DIW inks to determine if material properties can be used to predict the printability of a given ink [25,26]. Since DIW inks are complex fluids, rheology is often the technique of choice for measuring material properties of ink formulations. Many material properties have been investigated to predict printability, such as viscosity, modulus, and/or yield stress [27,28]. However, these metrics are either not universal across different fillers or are impossible to accurately measure on conventional rotational rheometers [29].

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Our recent research has shown that one method of accurately predicting printability that is agnostic to filler type is with the maximum packing fraction of the filler in the suspending base. This criterion predicts that inks are printable when particle loading is between 90 % and 94 % of the maximum packing fraction determined via experimental rheological characterization [29]. Though the normalized packing fraction of printable inks falls in a narrow range, the various inks formulations have particle loadings ranging from 50 % up to as high as 72 % depending on the filler particle due to differences in the maximum packing fraction. Here we expand on that result to validate the generality of the normalized packing fraction criterion by testing printability of 10 new particle fillers including both monomodal and multimodal blends of particles with a broader range of filler characteristics.

Many DIW inks use commercially available particle fillers which have a range of shapes and sizes in a binder. However, many theoretical and simulation studies are focused on hard spheres. It is a well-known fact that mono-disperse hard spheres have a random close packing fraction of 0.64 [30,31], which is the densest arrangement of non-overlapping spheres in a disordered state. In the context of suspension viscosity, the relationship between filler loading and the flow behavior of suspensions has long been a topic of significant interest [32–36]. The viscosity of a suspension is known to increase dramatically as the filler concentration approaches the maximum packing fraction. The maximum packing fraction can be extrapolated from experimental data; therefore, viscosity is a useful metric when characterizing these suspensions.

While hard sphere assumptions are necessary for many theoretical and modeling purposes [37], experimental approaches allow for measurements that can be correlated to filler characteristics that can be directly measured and related to maximum packing. The viscosity of suspensions is influenced by geometric factors beyond simple spheres, such as dispersity [38–40], aspect ratio [41–43], or surface roughness [44–46]. In the most general sense, dispersity is a measure of the breadth of a distribution of sizes for a given filler. Studies have shown that systems with larger dispersity, can achieve higher packing fractions than monodisperse systems. This is because smaller particles can fill the interstices between larger particles, leading to more efficient packing for the filler [47,48]. Thus, a polydisperse particle suspension will have a lower viscosity than a monodisperse suspension at the same particle loading. While some recent works have shown DIW in materials such as which take advantage of chain entanglement and/or chemical interactions such as hydrogen bonding [70], this work focuses on inks with frictional interactions.

Aspect ratio, the ratio of a particle's length to its width, is another critical factor influencing packing density and suspension viscosity [42,43]. Non-spherical fillers, such as rods, ellipsoids, platelets, and even fibers exhibit drastically different packing behaviors compared to their spherical counterparts [49–52]. Theoretical and experimental studies have shown that monodisperse systems with high aspect ratios such as rods can form more complex and often more efficient packing structures [53]. One well-known study shows that ellipsoidal M&M's with an aspect ratio of approximately 1.4 have a larger maximum packing fraction compared to spheres [54]. Since controlling aspect ratio in a filler can be difficult, there are few studies which have investigated this relationship experimentally for polydisperse systems. Meanwhile, recent studies have found different ways to define and control the filler roughness and found relationships between roughness and flow behavior of dense suspensions [45,55].

Despite the advancements in understanding the effects of dispersity and aspect ratio on dense suspensions independently the combined effects in AM inks with hard particle fillers is not well understood. From the experimental side, it is even more difficult, because controlling the dispersity and aspect ratio in a systematic fashion is a daunting task. However, one can correlate filler characteristics to packing density by utilizing a variety of commercially available fillers having a range of particle material properties.

In this study, we first validate the printability criterion laid out in our previous work with new DIW ink formulations. These results show that maximum packing fraction is an excellent predictor of DIW ink printability confirming that normalized packing fraction allows for predictions of printable ink formulations to be made. Next, we explore the impact of dispersity and aspect ratio on the maximum packing fraction and thus the printability of DIW inks. We provide an empirical relationship to predict the maximum packing fraction from filler microscopy alone, with an excellent fitting and validation correlation coefficients. Our findings provide a comprehensive understanding of how dispersity and aspect ratio influence DIW ink characteristics, paving the way for more efficient and effective DIW additive manufacturing formulation processes.

2. Materials and methods

Direct-Ink Write formulations were created using the silicone base (Non curing Part A, Sylgard™182, Dow Chemical $\rho = 1.03$ g/cm^3 [56]), and ten different fillers having varying morphologies and size distributions. The fillers used in this work are summarized in Table 1. Additionally, we studied blends of fillers to explore a greater variety of particle size distributions and dispersity including 1:1 mass ratio coarse to fine silicon dioxide (1:1 C:F SiO_2,), 3:1 ratio coarse to fine silicon dioxide (3:1 C:F SiO_2,) and 3:1 ratio of coarse to fine melamine (3:1 C:F Melamine).

Each filler was analyzed on the Morphologi 4-ID (Malvern Panalytical). The particle filler was dispersed onto a quartz plate using the sample dispersion unit under a pressure of 1 bar using dry nitrogen. Images were acquired for individual particles, and size and shape were quantified through the built-in software. Examples of particle images that were included in the analysis, and examples which were filtered out are shown in Figures SM1 and SM2**.** Multiple filler characteristics are calculated including particle size and aspect ratio distributions which will be discussed in detail in Section 5.

Each DIW ink was prepared in a planetary mixer (THINKY ARE-310), first by adding the non-curing silicone base and the filler. This sample was mixed at 2000 RPM for 2 min. A yttria-stabilized zirconia ball bearing of 1 cm in diameter was added after the first mix to ensure homogeneous mixing. These samples were then mixed for an additional eight minutes at 2000 RPM in two-minute intervals with hand mixing between each cycle. While mixing led to an increased temperature, all inks were allowed to cool before any printing or rheological measurements. Immediately prior to rheological testing, samples were mixed for two minutes to ensure a homogenous consistency. For inks in which two sizes of filler were used, both filler sizes were added to the base before any mixing occurred. No settling was observed on the timescale of the rheological experiments performed, nor the timescale of DIW printing.

The definition of printability is based on our previous work [29]. An ink is classified as being 'printable' if it satisfies two criteria: 1) the ink must be able to be extruded out of a syringe with a modest nozzle diameter and flow rate (0.84 mm nozzle at 0.25 mL/min) without exceeding a reasonable extrusion force (50 kgf or 6.1 MPa for a 10cc EFD syringe), and 2) the height of a printed cone must change less than 10 % over one minute after printing has been concluded. If an ink cannot

Table 1
List of individual fillers used in this work, including common names, shorthand names, manufacturer, density, and a reference to the SDS.

Filler	Shorthand Name	Manufacturer	Density (g cm^{-3})
Aluminum Oxide [57]	Al_2O_3	Almatis	3.95
Aluminum Nitride [58]	AlN	MSE Suppliers	3.25
Melamine [59]	Coarse Melamine	UL Prospector	1.57
Melafine [59]	Melafine	UL Prospector	1.57
Pentaerythritol [60]	PENTEK	Sigma-Aldrich	1.40
Silicon Dioxide [61]	Coarse SiO_2	US Nano	2.40
Silicon Dioxide [62]	Fine SiO_2,	US Nano	2.40

satisfy criterion 1) we say it is 'jammed'. If an ink cannot satisfy criterion 2) we say it is 'runny'. If an ink can satisfy both criteria, the ink is 'printable'. In each case, the inks were printed for five minutes, for a total ink volume of 1.25 mL. Examples of inks which were runny and printable can be seen in Figure SM3.

Rheology experiments were performed on a stress-controlled Discovery Hybrid Rheometer 3 (TA Waters) with 40 mm profiled/cross-hatched parallel plates. Prior to any experiment, the rheometer and test fixtures were calibrated for inertia and friction and the gap was zeroed. For all tests, DIW inks were loaded, and the gap was lowered to 1.05 mm, the top head was locked, and any excess ink was trimmed. If the material induced axial stresses which were too large for the instrument, a large-amplitude oscillation was applied while lowering the gap to the trim height. After trimming, the geometry was lowered to 1.00 mm under zero shear stress. All samples were held at zero shear stress for 5 min after the final gap was reached. Small amplitude oscillatory shear experiments were performed to measure the storage and loss moduli at a frequency of 1 Hz for 10 minutes at a constant stress amplitude, corresponding to a measured strain amplitude of less than 0.01 % [29]. For measured strain amplitudes less than 0.01 %, the moduli measured were independent of the applied stress Each test only requires fifteen minutes, opening the door for rapid formulation screening of DIW inks.

Statistical analysis was performed using Minitab (Minitab LLC). Regression tools were utilized to determine empirical fits, and which variables did not hold statistical significance. Validation tests were run on the final empirical model and are discussed further in Section 5.

3. Validation of proposed printability criterion

In our previous study, we discovered three criteria that separated three DIW inks into categories of 'runny', 'printable' and 'jammed'. In this study, we discuss the normalized packing fraction, while results testing the remaining criteria which were not as successful are discussed in the Supplementary Material. Based on our previous work, we predict that these DIW inks will be printable if the volume fraction is between 90 % and 94 % of the maximum packing fraction for a given filler. To investigate the predictive power of the proposed printability criterion, we first must determine the theoretical maximum packing fraction for each ink. A common way of determining maximum packing fraction is to plot the relative viscosity against volume fraction, and fit common filler packing models, such as Krieger-Dougherty or Mendoza [35,36].

An alternative method to calculate the maximum packing fraction utilizes a theory for packing of hard sphere which predicts that the relative viscosity will increase as volume fraction to the five-halves power ($\phi^{5/2}$). Thus, it is also possible to calculate the maximum packing fraction from a simple linear fit of the relative viscosity to the $-2/5$ (or -0.4) power versus packing fraction. The relative viscosity will diverge to infinity approaching the maximum packing fraction making it challenging to fit directly so the exponential rescaling can improve the quality of the fit. In our previous work, we found that the same scaling can be applied to the relative loss modulus which is more accurate to measure experimentally on suspensions close to their maximum packing fraction [29]. While the complex viscosity would seem like the more obvious choice, that material property would be dominated by elastic energy storage ($G' \gg G'' \Rightarrow \eta^* \approx O(G')$) and thus would not be representative of dissipation in the suspension. Therefore, we proposed utilizing the relative loss modulus (G''_r) to determine the maximum packing fraction.

$$G''_r(\phi) = \frac{G''_{SAOS}}{G''_0}, \tag{1}$$

where G''_r is the relative loss modulus, G''_{SAOS} is the experimental measurement of the loss modulus at small stresses or strains, and G''_0 is the loss modulus of the resin base. For the exemplar resin used in this study, Sylgard™182 [56], the kinematic viscosity of the base is 5639 cSt at 25 °C, or a loss modulus $G''_0 = \eta_0\omega = 34.4$ Pa at 1 Hz (2π rad/s). Instead of choosing one of the many possible models to fit the data, the methods of Tanner and Pal to estimate the maximum packing fraction, using the loss modulus by plotting the reduced loss modulus to the power of -0.4, coming from the reciprocal of the power or 2.5 common to many particle fitting models [29,68,69]. Fig. 1 shows $G''^{-0.4}_r$ vs volume fraction ϕ for the 10 new particle fillers studied here. The x-axis intercept for each filler represents the maximum packing fraction extrapolated from the experimental data. The linear trend observed for all the fillers plotted in Fig. 1 validates the use of this exponential rescaling. Due to the variety of particle size distributions and morphologies, there is a wide range of maximum packing fractions from 56 – 78 vol%, which will be discussed quantitatively in Section 4. The raw data used in this analysis can be found in Table SM1.

In our previous study, three fillers not included here had inks which were printable when they had filler loadings between 90 % and 94 % of the maximum packing fraction. To see if this printability range is predictive, we see if the printability of these new fillers falls in the same printability window [29]. This gives a testable hypothesis of which ink formulation(s) in this new set will be printable. Each ink was printed as discussed above in Section 2 and our previous work [29], and the results are shown in Fig. 2 by filler type. The dark blue box for each filler is the region of 90–94 % of the maximum packing fraction calculated from Fig. 1. The symbols show the results of printing tests with inks formulated with a range of particle loadings. The red diamonds represent inks which were 'runny' and could not hold their shape, the blue diamonds represent inks which were 'printable', and the black diamonds represent inks which were 'jammed' and could not be printed. Every ink with a normalized particle loading between 90 % and 94 % of maximum packing was indeed 'printable' as predicted. Moreover, no inks deemed 'runny' or 'jammed' fell into this predicted printability window, validating the predictive power of the normalized packing fraction printability criterion. Fig. 3 shows box plots for all the fillers, by showing the normalized volume fraction on the y-axis. Again, the dark blue box shows the predicted printability range of 90–94 % of the maximum packing fraction. From this normalization, we see the power of this criterion even more clearly. These validation experiments provide robustness to the previous predictions and solidify that maximum packing fraction is a crucial factor in determining the printability of a

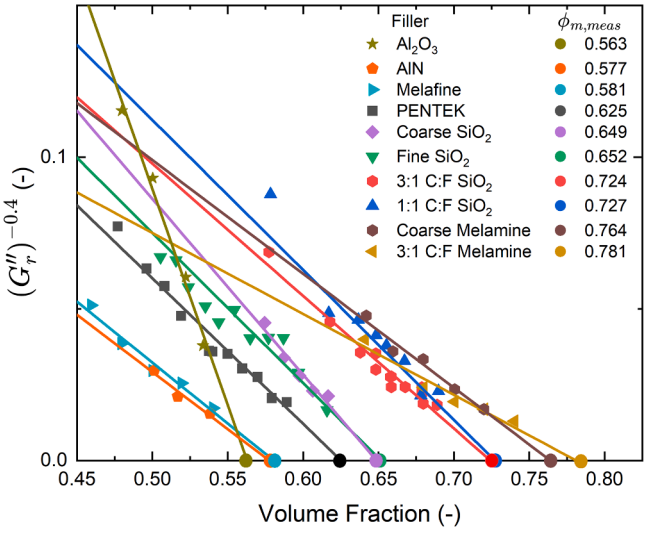


Fig. 1. Relative loss modulus raised to the negative two-fifths (-0.4) power plotted against volume fraction for each of the ten particle fillers, with lines showing the linear fit. The intersection of the fitting line with the x-axis is the maximum packing fraction, indicated by a large circle. The maximum packing fractions for each filler are listed in the legend.

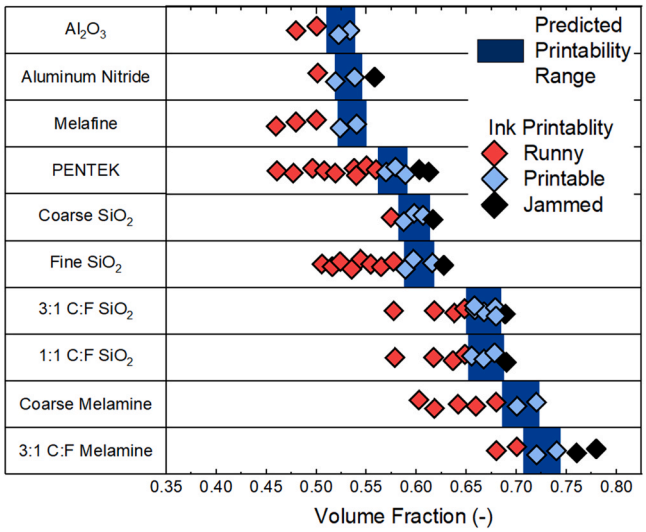


Fig. 2. Scatter plots for inks with different particle fillers as a function of volume fraction on the x-axis. Red diamonds show inks which are 'runny', blue diamonds show inks which are 'printable' and black diamonds show inks which are 'jammed'. The blue shaded region shows the printability range of normalized packing fraction between 90 % and 94 % from previous work [29]. All printable inks tested fell into this range of normalized packing fraction.

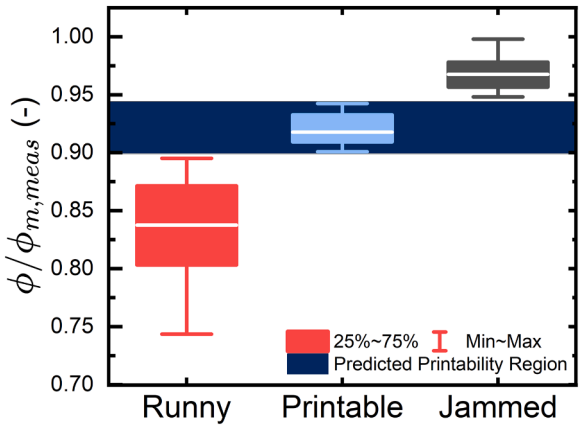


Fig. 3. Box plots for DIW ink formulations presented in this paper showing packing fraction (\phi) normalized by the extrapolated maximum (\phi_{m, meas}) as a function of ink printability. Red box plot shows inks which are 'runny', blue box plot shows inks which are 'printable' and black box plot shows inks which are 'jammed'. The dark blue box shows the predicted region of printability based on normalized packing fraction from our previous work on particles not included in the above data [29].

DIW ink formulation for a wide variety of hard particle fillers.

This methodology of utilizing small-amplitude oscillatory shear to screen many inks and estimate maximum packing fraction can be performed in a few hours. Printing many inks and determining the printability through post-mortem analysis, meanwhile, can be a process requiring days to complete. This faster screening also allows for better optimization of ink as limited resources can be focused on a narrow window of ink formulations. These rheology methods provide an easy quality check on ink formulations as the linear plateau properties will be very sensitive to changes in filler characteristics from different lots or batches.

4. Predicting maximum packing fraction from filler characteristics

Since DIW ink printability is clearly related to the maximum packing fraction of a given filler, the next step is to determine if measurable particle filler attributes can be used to predict the maximum packing fraction without the need to mix multiple ink formulations for testing. As discussed in the introduction, dispersity, aspect ratio, and surface roughness are filler characteristics which have been studied in relation to maximum packing fraction. When dealing with dense suspensions such as DIW inks, where the control over filler morphology may be limited, we aim to determine which filler attributes have the largest effect on maximum packing fraction for fillers as-received. In measurements of the fillers through microscopy, we obtain numerous potential metrics, including circle equivalent (CE) diameters, length and width, aspect ratio, circularity, convexity, and solidity. When determining which filler attributes may affect maximum packing fraction, we initially started with all the particle properties. We found that circularity, convexity, and solidity did not vary much amongst the various filler types and the variations did not appear to be correlated with printability. The list of measured filler attributes which were not found to be correlated with maximum packing fraction are shown in Table SM2. For the 2-D particle microscopy utilized there were no good measures of particle roughness which correlated to the maximum packing fraction. The most significant properties were dispersity and mean aspect ratio.

While the CE diameter did not correlate directly with maximum packing, the dispersity of particle sizes did. When defining dispersity, the volume-weighted (De Brouckere mean, $d_{4,3}$), and surface-area weighted (Sauter mean, $d_{3,2}$) mean CE diameters are used to calculate the spread of a log normal distribution, where the volume-weighted and surface-area weighted CE diameters are higher order moments of the number-weighted average. The filler dispersity has been calculated as $\sigma^2 = \ln(d_{4,3}/d_{3,2})$ [63,64], where larger values correspond to samples which have a larger spread of sizes and thus larger dispersity. Particle size distributions for all the fillers used in this work are shown in Fig. 4, where each filler has a vertical dashed line representing the surface-area weighted mean size, and a dotted line for the volume-weighted mean size.

The aspect ratio for each particle in a filler is the longest length divided by the widest width of a particle, chosen such that it is greater than one. Table 2 shows the dispersity (σ^2) and arithmetic mean aspect ratio (AR) for each filler used in this study sorted by increasing maximum packing fraction in the same way as Fig. 1, along with the fillers used in our previous work shown in the last three rows. The dispersities of each filler ranged from 0.114 to 0.637, and the mean aspect ratios ranged from 1.07 to 1.44. Even fillers which were sold as 99 % spherical had some dispersity greater than zero, and aspect ratios greater than one. Thus, it is important to measure these filler characteristics for each new batch, so that behaviors of the corresponding DIW ink can be understood and predicted.

With these measured filler attributes, and maximum packing fractions, analysis can be performed to determine maximum packing fraction from filler measurements alone. In this work we have presented data on ten new fillers, along with the three fillers from our prior work [29]. Utilizing these thirteen formulations, we developed a quadratic model to estimate maximum packing fraction as a function of dispersity and aspect ratio, while utilizing a 5-fold cross validation method [65]. This validation method separates the data into five groups, and performs a regression on four of the five groups, while holding the final group as a test set. This process is repeated, where each group gets used as a test set, while the other sets are used for fitting. The final regression coefficients are the weighted average of the five runs. This led to the following empirical relationship to predict the maximum packing fraction as a function of dispersity and aspect ratio,

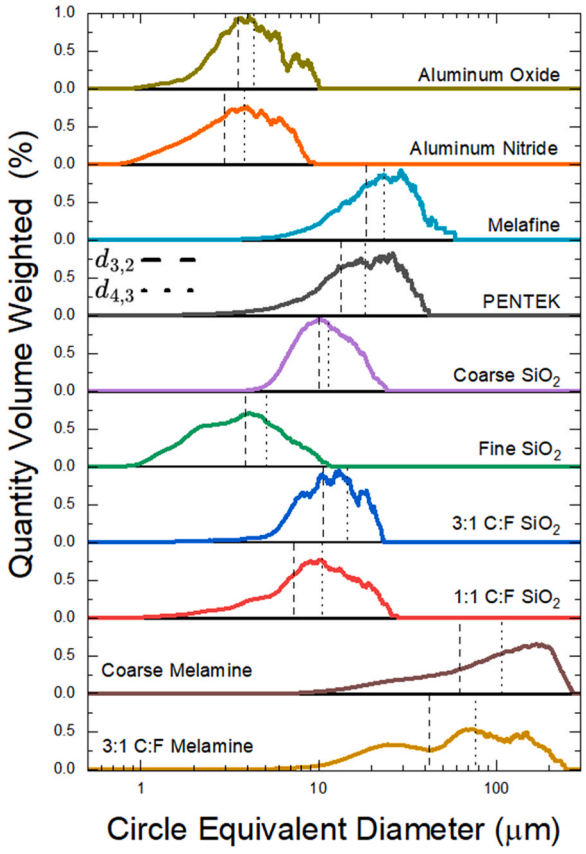


Fig. 4. Particle size distributions for all fillers used in this work, sorted from smallest maximum packing fraction at the top to largest maximum packing fraction at the bottom. Each panel contains two vertical lines, the dashed left line represents the surface-area weighted mean $d_{3,2}$, while the right dotted line represents the volume-weighted mean $d_{4,3}$.

$$\phi_{m,FCM} = 5.95 + 1.897\sigma^2 - 9.04\mathrm{AR} + 0.407\left(\sigma^2\right)^2 + 3.756\mathrm{AR}^2 - 1.409\mathrm{AR}\sigma^2 \quad (2)$$

with a fitting R^2 of 98.5 %, and a validation R^2 of 92.1 %. The impact of each factor can be elucidated by examining the standardized coefficients, which are shown in Figure SM6, and a Pareto chart for significance, shown in Figure SM7. In this case, dispersity has the strongest effect on the maximum packing fraction, followed by aspect ratio, and then the square term of the aspect ratio. A positive coefficient indicates that the particle attribute is positively correlated with maximum packing fraction.

The results of the filler characteristics model are shown as a contour plot in Fig. 5, where the lighter regions represent smaller maximum packing fractions, and the darker regions represent larger maximum packing fractions. Contour lines are shown in increments of 0.1 for maximum packing fractions starting at 0.55 and ending at 0.95. Each of the filler types is plotted according to their measured dispersity and aspect ratio. The shapes representing each filler correspond to the raw error between the model prediction for maximum packing fraction, with the maximum error being less than 0.02.

The first test for this model is how well does it predict maximum packing for monodisperse hard spheres which corresponds to a dispersity of zero, and an aspect ratio of one and an expected maximum packing fraction of 0.64 [31]. Our filler characteristic model predicts a maximum packing fraction of 0.66 which is a close match to the

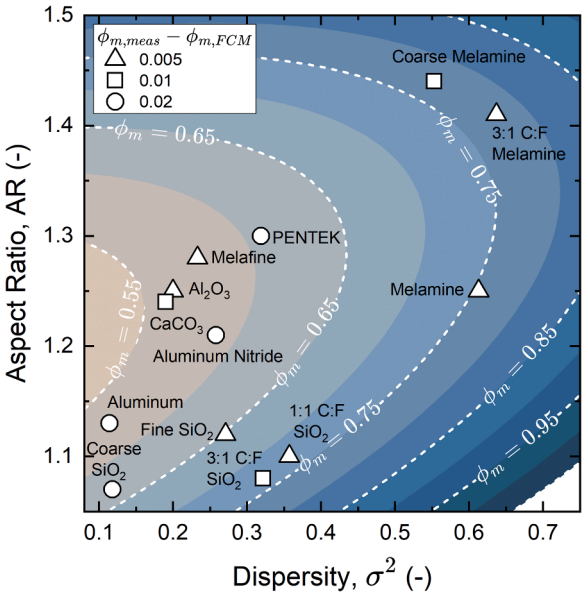


Fig. 5. Contour plot comparing experimental measurements (symbols) and predictions of the filler characteristics model (underlying contour plot) as a function of particle dispersity (\sigma^{2}) and aspect ratio (AR) . White dashed isolines show the predicted maximum packing fraction. Three types of markers show where the experimental data lie in the dispersity vs. aspect ratio space: triangles, squares, and circles represent the fillers where the predicted maximum packing fraction from the filler characteristics model is within 0.005 (0.5 vol%), 0.010 (1.0 vol%), and 0.020 (2.0 vol%) respectively, when compared to the measured value of maximum packing from the rheological fitting.

Table 2
Filler characteristics including surface-area and volume-weighted mean particle size, dispersity and aspect ratio for each filler measured on the Morphologi 4 ID. Also shown are maximum packing fraction measured with rheology $\phi_{m,meas}$ predicted maximum packing fraction from the filler characteristics model $\phi_{m,FCM}$ and the absolute difference, in volume percent. The last three rows contain particles which were utilized in our previous work [29]

Filler	$d_{3,2}$ (μm)	$d_{4,3}$ (μm)	σ^2 (-)	AR (-)	$\phi_{m,meas}$ (-)	$\phi_{m,FCM}$ (-)	$\phi_{m,meas} - \phi_{m,FCM}$ (vol%)
Al_2O_3	3.56	4.35	0.200	1.25	0.563	0.562	0.0822
Aluminum Nitride	2.95	3.82	0.258	1.21	0.577	0.587	−1.044
Melafine	18.53	23.40	0.233	1.28	0.581	0.577	0.4491
PENTEK	13.33	18.33	0.319	1.30	0.625	0.608	1.71
Coarse SiO_2	10.1	11.36	0.118	1.07	0.649	0.629	1.99
Fine SiO_2	3.88	5.09	0.271	1.12	0.652	0.653	−0.104
3:1 C:F SiO_2	10.61	14.65	0.322	1.08	0.724	0.731	−0.683
1:1 C:F SiO_2	7.28	10.41	0.357	1.10	0.727	0.727	0.0453
Coarse Melamine	62.22	108.2	0.553	1.44	0.764	0.772	−0.833
3:1 C:F Melamine	41.78	79.03	0.637	1.41	0.781	0.779	0.208
Aluminum [29]	13.21	14.81	0.114	1.13	0.552	0.571	−1.89
$CaCO_3$ [29]	7.89	9.54	0.190	1.24	0.553	0.559	−0.581
Melamine [29]	35.11	64.79	0.613	1.25	0.757	0.755	0.210

theorical value and is within the error seen for all our filler measurements of ±0.02. Fig. 5 shows the same strong positive relationship between dispersity and maximum packing fraction in qualitative agreement with literature [30,38,40].

The relationship between maximum packing fraction and aspect ratio is more complex. For a constant dispersity, an increase in the aspect ratio corresponds to a decrease in the maximum packing fraction for small aspect ratios. However, as aspect ratio continues to increase, the maximum packing fraction reaches a minimum before starting to increase. The exact aspect ratio for this transition depends on the dispersity due to the cross term of aspect ratio times dispersity in Eq. 2. This is different than the theoretical and simulation results seen in literature where the most common understanding is that maximum packing fraction will increase with increasing aspect ratio, followed by a decrease after aspect ratios reach 1.5. [53,54]. Recent experimental work has suggested that that the maximum packing fraction for fillers with a distribution of sizes and aspect ratios decreases when the aspect ratio is greater than one [44,66,67].

While not comprehensive, we believe the filler characteristics model captures the observed response over a range of filler dispersities and aspect ratios which will encompass many of the real-world fillers commonly used in DIW inks. This model can be used to predict the maximum packing fraction of a DIW ink based on the characteristics of the filler alone. Now we must determine how well this empirical relationship predicts the region of printability for DIW inks.

5. Printability predictions from filler characteristics model

In Section 3, we validated the printability criterion based on the normalized packing fraction as determined from fitting the rheological response of ten new DIW inks. Meanwhile, in Section 4, we introduce an empirical relationship to determine maximum packing fraction based only on measured filler attributes. Fig. 6 compares experimental measurements of printability (symbols) and the printability prediction using the filler characteristics model in Eq. 2 (patterned boxes) against the actual printability range shaded in dark blue. The symbols show all the inks tested in this work and our previous work [29], separated by color

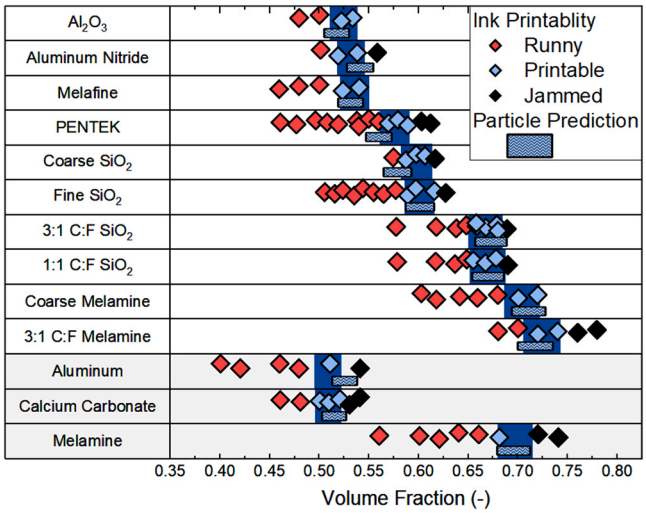


Fig. 6. Scatter plots for inks with different particle fillers, including inks from our previous work, as a function of volume fraction on the x-axis. Red diamonds show inks which are 'runny', blue diamonds show inks which are 'printable' and black diamonds show inks which are 'jammed'. The dark blue shaded region shows the printability range of normalized packing fraction between 90 % and 94 %. The blue box plots with infill show the printability region predicted from the filler characteristics model and measured filler dispersity and aspect ratio. The filler characteristcs model predicts the printability region within a volume fraction of 0.02 for all fillers.

into 'runny', 'printable', and 'jammed'. As discussed in Section 3, all the measured inks which were printable fell within the range of printability predicted by the measured rheological characteristics (shaded dark blue), validating the robustness of this measure to predict printability. The empirical estimate of the printability region based upon the measured filler characteristics is within 0.02 (2 vol%) of the measured printability range for all the DIW inks. This result means that if one can measure the dispersity and aspect ratio of their filler, then they can predict the range of particle loadings for printable DIW inks within two volume percent, *prior to formulating any inks*.

6. Conclusion

In our previous work, we established a relationship between maximum packing fraction and printability of Direct Ink Write (DIW) inks, providing a framework to quantify printability through small-amplitude oscillatory shear measurements and maximum packing fraction [29]. We proposed a criterion that printable DIW inks would have normalized packing fractions between 90 % and 94 %, based on three different fillers. In this work, we tested and validated the predictions of our previous criterion [29] on ten new particle fillers and established that the maximum packing fraction is a critical parameter in determining the printability of DIW inks. Depending on the particle, the printable inks have particle loadings that vary from 50 to 78 vol%, but fall between a normalized packing fraction of 90–94 %. Measurement of maximum packing fraction only requires a handful of simple small amplitude oscillatory shear measurements and can be completed in an hour or two for a new particle filler as compared to several hours to print many possible ink formulations. This enables better optimization of ink formulations as the best range of particle loadings can be identified early.

Building on the foundation of normalized packing fraction, we introduce an empirical filler characteristics model capable of estimating the maximum packing fraction of fillers from microscopy alone, based on dispersity and aspect ratio. The data were used to formulate a second order model using these two parameters, with an R^2 of 92.1 % based on 5-fold validation testing. This advancement significantly enhances the predictive power and practical utility of our normalized packing fraction framework. By incorporating filler dispersity and aspect ratio into our model, we can now predict the maximum packing fraction, and hence the filler loading range, that will result in a printable ink before the first ink formulation has been mixed, thereby further streamlining the formulation process.

Traditionally, development of new DIW inks has relied on an iterative print and test approach only by varying volume fractions of fillers. Our findings demonstrate that printability can also be controlled by adjusting filler dispersity and aspect ratio. Strategies for increasing the maximum packing fraction allows DIW users to increase the filler loading without compromising the printability of the ink formulation. This has profound implications for various applications, such as increasing energy density in batteries, creating stronger dental composites, or development of more robust concretes.

We have demonstrated the utility of this technique for a broad range of hard particle fillers in a silicone binder. The maximum packing fraction captures dependencies on particle filler morphology and size distributions, which allows for direct comparisons when used as a normalization factor. This work will also be useful for fields of DIW which may want inks to spread to fill a mold, for example. In fact, the specific ranges can be determined for a given use case, and the filler characteristics model will determine which range of filler loadings will give the best results for any new filler.

This work is applicable to any DIW inks in which frictional interactions are dominate. The printability criterion may not be suitable for all types of filler such as attractive colloids, where structure is induced by surface chemistry rather than frictional jamming. Similarly, the methodology may not apply to deformable, flexible, or high aspect

ratio fillers. Maximum packing fraction could also depend on the choice of binder or the use of additives such as dispersants. This framework does provide a more efficient methodology for quickly assessing the impact of formulation changes without time consuming iterative testing of new inks.

In summary, this work provides the DIW practitioners with a robust method for determining ink printability based on the maximum packing fraction of each hard particle filler. Additionally, we propose a new model to estimate maximum packing fraction based solely on filler characteristics including the filler dispersity and aspect ratio. The range of filler loadings which are printable can be determined through small-amplitude oscillatory shear experiments or estimated closely by performing microscopy or another characterization technique on the filler size and shape and using our empirical relationship above, both of which are significantly more time efficient than traditional approaches. Combined, this theory of printability opens the door for rapid prin,tability screenings across various formulations, filler types, morphologies, and size distributions to enable optimized DIW material extrusion ink formulation.

CRediT authorship contribution statement

Jessica W. Kopatz: Writing – review & editing, Writing – original draft, Resources, Conceptualization. **Anne M Grillet:** Writing – review & editing, Resources, Conceptualization. **Simon A Rogers:** Writing – review & editing, Resources. **Alexander S Tappan:** Writing – review & editing, Funding acquisition, Conceptualization. **James J Griebler:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.addma.2025.104651.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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